



BioLinkerAI: Capturing Knowledge Using LLMs to Enhance Biomedical Entity Linking

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Abstract. In this paper, we introduce *BioLinkerAI*, a neuro-symbolic approach tailored for biomedical entity linking. Traditional domain-specific entity linking approaches necessitate substantial labeled-training datasets and present challenges when adapting to each new dataset or setting. *BioLinkerAI* integrates symbolic methodologies with sub-symbolic models to mitigate the constraints of limited training data availability. The symbolic component utilizes a rule-based entity extraction mechanism, underpinned by an extensive set of linguistic and domain-specific rules. Concurrently, the sub-symbolic component employs a Large Language Model (LLM) to achieve precise candidate disambiguation. This mechanism enhances entity linking accuracy, especially when a single entity in a knowledge base such as UMLS aligns with multiple terms, leveraging the contextual intelligence encapsulated within the LLM's embeddings. Empirical evaluations conducted on a range of biomedical benchmarks demonstrate the superior performance of *BioLinkerAI*. Notably, it surpasses existing benchmarks in entity linking accuracy (e.g., achieving an improvement from **65.4** (best baseline) to **78.5** (our model) on unseen data accuracy, representing the most stringent evaluation paradigm). Additionally, *BioLinkerAI* consistently performs adeptly on both structured sentences and individual keywords. To facilitate broader utilization, the source code, datasets, and a public API are available (<https://github.com/SDM-TIB/BioLinkerAI>).

Keywords: Biomedical entity linking · Neuro-symbolic approach · Large Language Model · Rule-based entity extraction · Contextual intelligence

1 Introduction

In today's digital age, data availability has surged dramatically, with similar trends expected to continue. While much of this data is structured (e.g., relational databases, Knowledge Graphs (KGs)/Bases (KBs)), a large amount remains unstructured. The challenge of connecting unstructured and (semi-)structured information has garnered considerable scholarly interest. Entity linking (EL) is key in associating textual entities with their counterparts in a KG or KB, extensively studied in literature, especially in domains like question answering and social media [17]. One main challenge in EL is resolving ambiguity in text [6], particularly in specialized fields like biomedicine,

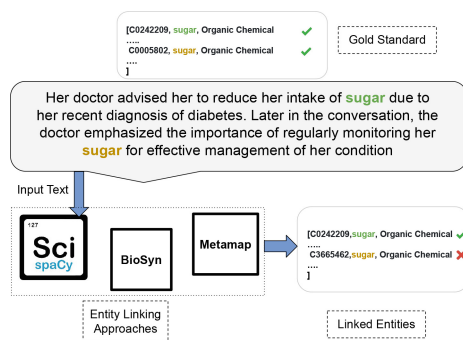


Fig. 1. Complexities in Biomedical Entity Linking. The term "sugar" appears in distinct contexts, yet shares an identical label within UMLS. Resolving such ambiguities necessitates specialized domain knowledge, such as detailed entity descriptions.

where domain knowledge is crucial [2]. For example, in Fig. 1, the surface form "sugar" must be linked to distinct concepts in the Unified Medical Language System (UMLS) KB, both sharing the same label. Biomedical EL approaches often struggle with such cases, but large language models (LLMs) like GPT-4 can re-rank candidates using domain-specific knowledge¹. However, LLMs alone cannot perform end-to-end EL². Our work, therefore, leverages LLMs for disambiguation alongside symbolic knowledge for biomedical entity linking. **Proposed Approach and Contributions** In this work, we aim to discard the necessity of (1) the availability of annotated gold samples, and (2) training large machine learning models on labeled data for biomedical entity linking over new datasets. We propose *BioLinkerAI*, a Neuro-Symbolic approach that leverages symbolic and sub-symbolic components. The symbolic components exclude the requirement of availability of labeled training data and rely purely on a catalog of linguistics and domain-specific rules. Considering that most biomedical domain entities have definitions or other metadata from various sources can contribute to overcoming the limitation of available training data. Consequently, the integration of a pre-trained sub-symbolic element augments the precision of the symbolic constituents by leveraging the information embedded within the metadata associated with medical entities; it relies on the entities' definition/description to rank the list of candidates generated by the symbolic component. Furthermore, our approach can be generalized to other KBs containing biomedical entities. We make the following key contributions: **1) BioLinkerAI**, a pioneering neuro-symbolic method for EL within the biomedical field, which performs efficiently with both sentences and keywords. The performance of *BioLinkerAI* is studies over biomedical benchmarks, showing that it significantly surpasses current SOTA EL techniques (from **65.4** (best baseline) to **78.5** (BioLinkerAI) on previously unseen data, the most difficult scenario). **2) A semantic-based model** that captures the semantics of an entity expressed within a text. We establish a background knowledge to characterize entities from various knowledge graphs. **3) A rule-governed framework**

¹ <https://chat.openai.com/share/094cda83-be0f-41a4-87ad-956470e71ef9>.

² <https://chat.openai.com/share/1678adcc-5af9-48de-8ef7-9344c2612a63>.

engineered to embody the domain-specific knowledge crucial for identifying and associating entities within specialized fields (e.g., medical). We formulate guidelines to capture domain-specific characteristics and linguistic features of entities. The paper’s structure is as follows: the next Section summarizes the related work. Section 3 describes the problem statement and approach. Section 4 explains the experimental settings followed by results. We conclude in Sect. 5.

2 Related Work

Entity Linking (EL) over Knowledge Bases. The field of entity linking has been thoroughly investigated, with methodologies spanning from graph-based algorithms [21] to approaches utilizing neural networks [12]. A generative framework that eschews hard negative sampling in the training phase was proposed by [5]. These techniques bolster EL efficacy by incorporating supplementary data, including entity descriptions, categorizations, and KG triplets, to facilitate the learning process [3]. Numerous models in this domain operate on predetermined candidate sets, frame the task as a classification challenge, and typically necessitate substantial annotated training corpora. To mitigate reliance on labeled datasets, strategies like those presented in [7] exploit unannotated information sources. In recent developments, zero-shot paradigms have emerged, expanding EL capabilities to encompass entities absent from the training data [11].

Biomedical Entity Linking is an active area of research. [9] add a softmax classification layer, while other works map surface forms and entity candidates into a shared space, disambiguating candidates through nearest-neighbor methods [4, 24]. [23] propose a generative model that incorporates synonyms and definitions via KG-guided pre-training. Approaches such as [2, 22] follow a retrieve-and-rank strategy, with [22] introducing cross-domain data integration to transfer knowledge from general text KGs to the biomedical domain. In summary, the field is rich with various biomedical EL techniques. Unlike previous methods, our approach leverages the closed-form structure of a KB (e.g., UMLS) to encode entity information, enhancing the performance of a symbolic component, inspired by human-defined rule templates [6, 16]. *BioLinkerAI* sets itself apart by eliminating the need for labeled data during inference, instead relying on linguistic rules to generate entity candidates.

3 The BioLinkerAI Approach

Problem Statement

The *BioLinkerAI* methodology aims to associate surface forms (alternatively termed mentions) from a given passage with their corresponding entity representations within a knowledge graph (KG). This association process employs a particular approach, which we can express mathematically as follows. In formal terms, we can denote a brief text as a collection of tokens $\mathcal{T} = \{t_1, \dots, t_n\}$. A KG comprises a set of resources $\mathcal{U}$. Let $\mathcal{P}(\mathcal{T})$ and $\mathcal{P}(\mathcal{U})$ signify the power sets of tokens and resources, respectively. The objective of associating tokens from a text to the power set of resources $\mathcal{P}(\mathcal{U})$ is characterized by a partial function ($p : \mathcal{P}(\mathcal{T}) \rightarrow \mathcal{P}(\mathcal{U})$). We define a function ϕ , serving as an ideal oracle

that flawlessly links a group of tokens to the appropriate resources/entities, expressed as $\phi : \mathcal{P}(\mathcal{T}) \rightarrow \mathcal{P}(\mathcal{U})$. Let $acc(.,.)$ denote a utility function that assesses the accuracy of the association generated by $p(.)$ relative to the oracle mapping $\phi(.)$. The function $p(.)$ is required to meet the following optimization criterion. **Accuracy** of the associations must be optimized: $\arg \max_{token \in \mathcal{P}(\mathcal{T})} acc(\phi(token), p(token))$.

Proposed Solution

We propose a neuro-symbolic approach, denoted as $p(.)$, integrating symbolic and sub-symbolic components. Formally, the symbolic component can be described as a function $\mathcal{S} : \mathcal{T} \rightarrow \mathcal{P}(\mathcal{U})$ that uses rules to map tokens from $\mathcal{T}$ to entities in $\mathcal{P}(\mathcal{U})$ in a KG. The function $\mathcal{S}$ is built on top of a knowledge base (e.g., UMLS), relying on a catalog of linguistic and domain-specific rules. The sub-symbolic component, denoted as $\mathcal{L} : (\mathcal{T}, \mathcal{P}(\mathcal{U}), \mathcal{B}) \rightarrow \mathcal{P}(\mathcal{U})$, handles the disambiguation of candidate entities. Here, $\mathcal{B}$ represents the background knowledge embedded in the descriptions and definitions of entities in the KG. $\mathcal{L}$, therefore, predicts the best candidates based on the context of the input text and the description of each candidate. The function $p(.)$ combines the symbolic and sub-symbolic components: $p(token) = \mathcal{L}(\mathcal{S}(token), \mathcal{B})$. The function $\mathcal{S}(token)$ extracts the entity candidates from the tokens, and the function $\mathcal{L}(\mathcal{S}(token), \mathcal{B})$ disambiguates these entities considering the context of tokens and descriptions of the candidates in $\mathcal{B}$. Therefore, our solution fulfills the conditions of accuracy, token maximization, and entity minimization outlined in the problem statement.

Symbolic Component - Background Knowledge: Every entity (resource) within the BK is characterized by labels, resourceID, semantic type, language, sameAs links to knowledge graphs, and a heuristically derived confidence metric that enumerates the quantity of shared labels for a resource across diverse knowledge graphs. The sameAs connection enriches the background knowledge by integrating information from KGs beyond the primary target (offering additional labels).

The neuro-symbolic methodology relies on the heuristic score for disambiguation, attributing higher values to resources exhibiting identical labels across various sources (stage 3 in Fig. 2). This background knowledge is structured as a deductive database, comprising both intensional and extensional components. The extensional component encompasses ground predicates that delineate the fundamental attributes of resources within the background knowledge, including semantic classifications, definitions, labels, and their originating providers (Fig. 2 - Background Knowledge).

A Rule-governed framework characterizes entities by their mentions (e.g., as illustrated in phase 1 of Fig. 2), employing a collection of domain-specific and linguistic rules to identify surface forms within the input natural text and associate them with the target knowledge graph.

Domain-specific Rules establish the criteria for entity identification within a particular field. As a conceptual demonstration, we harness the attributes of resources in a knowledge graph to define entities. Our guidelines operate on the premise that entities are distinguished by their definitions, semantic classifications, and labels. Considering that numerous knowledge graphs are community-maintained, a single resource may possess multiple values for any given attribute. Furthermore, a resource might have equivalent

to the original query. The enriched and ranked list of entity candidates is represented as $\mathcal{G}(\text{token}) = \mathcal{L}(\mathcal{R}(\mathcal{E}(\mathcal{S}(\text{token}))))$, $\mathcal{B}$) where $\mathcal{L}$ denotes the sub-symbolic component that apply further domain-specific rules for re-ranking.

In summary, the candidate generation process $\mathcal{G}$ integrates the initial retrieval of candidates based on string similarity through a search engine $\mathcal{S}$, enrichment of candidates $\mathcal{E}$, ranking using the BM25 algorithm $\mathcal{R}$, and potential re-ranking via domain-specific rules before passing the candidates to the sub-symbolic component $\mathcal{L}$.

Sub-symbolic Component - Contextual knowledge embedded within the background knowledge $\mathcal{B}$ is leveraged by the sub-symbolic component, denoted as $\mathcal{L} : (\mathcal{C}, \mathcal{B}) \rightarrow \mathcal{P}(\mathcal{U})$, for disambiguating candidates received from the previous step. For each candidate in $\mathcal{C}$, an ID, label, semantic type, and definition are provided to assist the sub-symbolic system in predicting the best candidate. Therefore, the operation of $\mathcal{L}$ can be formally expressed as follows: $\mathcal{L} : (\text{candidates}, \mathcal{B}) \rightarrow \mathcal{P}(\mathcal{U})$ where *candidates* $\in \mathcal{C}$ denotes the set of entity candidates obtained from the symbolic component. The function $\mathcal{L}$ leverages the contextual knowledge within $\mathcal{B}$ to select the most suitable entities.

A Large Language Model is responsible for the final disambiguation task, defined as a function $\mathcal{M} : (\mathcal{T}, \mathcal{P}(\mathcal{U}), \mathcal{B}) \rightarrow \mathcal{U}$, where $\mathcal{M}$ predicts the most probable entity given a text, list of candidates, and the properties of the candidates from the background knowledge: $\mathcal{M} : (\mathcal{T}, \mathcal{P}(\mathcal{U}), \mathcal{B}) \rightarrow \mathcal{U}$

$\mathcal{T}$ represents the input text, $\mathcal{P}(\mathcal{U})$ is the list of candidates for each token, and $\mathcal{B}$ signifies the background knowledge. The function $\mathcal{M}$ operates on a prompt, designed to specifically consider the description and semantic type of the candidates relative to the context of the input text. Providing such contextual knowledge helps the LLM in reducing ambiguity among the list of candidates and in identifying the most relevant candidate, effectively fulfilling the disambiguation process as outlined in the problem statement. Figure 2 illustrates this procedure, where the LLM disambiguates between candidates based on their properties. An example of the prompt is provided in the link³.

Table 1. |Entities| is the quantity of distinct entities. |Mentions| is the total count of mentions.

		MedMentions	BC5CDR
Mentions	Train	120K	18K
	Dev	40K	934
	Test	40K	10K
Entities	Train	19K	2K
	Dev	9K	281
	Test	8K	1K

Table 2. Ablation Study showing how symbolic and sub-symbolic components complement each other’s performance (accuracy).

	MedMentions	BC5CDR
BioLinkerAI wo/ sub-symbolic component	65.3	80.4
BioLinkerAI wo/ domain-specific rules	74.3	87.3
BioLinkerAI	81.3	93.3

³ <https://chat.openai.com/share/22c2619a-6709-4560-acb5-b40597fcea4c>.

Table 3. Accuracy using two settings. Baseline values are from [1,2]. For *BioLinkerAI*, unseen and seen scenarios yield similar results, as our approach refrains from training with entity-specific sentences. *BioLinkerAI* demonstrates a significant performance advantage over baselines. LLM in isolation can not perform entity linking.

	MedMentions			MedMentions w/ Gold Types			BC5CDR			B C5CDR w/ Gold Types		
	Overall	Acc. on		Overall	Acc. on		Overall	Acc. on		Overall	Acc. on	
	Acc.	Seen	Unseen	Acc.	Seen	Unseen	Acc.	Seen	Unseen	Acc.	Seen	Unseen
SciSpacy [13]	38.8	48.3	26.6	47.2	59.8	31	53.9	61.8	41.7	54.2	61.8	41.9
N-GRAM TF-IDF [14]	50.9	50.9	51	67.9	69	64	86.9	89.2	74.6	87.8	90.2	76.1
BIOSYN [19]	72.5	76.5	58.7	77	80.7	64.1	87.8	89	81.1	87.9	89.1	81.3
SAPBERT [10]	69.8	72.9	58.9	74.1	77	63.8	85.2	85.8	82	86	86.8	82
INDEPENDENT [11]	72.8	75.9	61.9	76.8	79.2	68.4	90.5	94	73.6	90.6	94.1	73.6
CLUSTERING-BASED [2]	74.1	77.3	62.9	79.1	81.5	70.5	91.3	94.9	73.8	91.4	94.9	74
DUAL [1]	75.7	79.9	60.9	NA	NA	NA	NA	NA	NA	NA	NA	NA
ARBORESCENCE [1]	75.7	80	61	NA	NA	NA	NA	NA	NA	NA	NA	NA
BioEFG [18]	76.7	79.9	65.4	NA	NA	NA	93.4	95.6	81.2	NA	NA	NA
[25]	66.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
BioLinkerAI-Llama2(70b)	79.4	81.4	76.1	82.1	81.6	78.2	93.1	95.2	90.1	93.3	95.2	89.8
BioLinkerAI-GPT4	81.3	83.4	78.5	84.1	83.6	79.5	93.3	95.4	90.2	93.8	95.6	90.7

4 Experiment and Results

The objective of our experiments is to evaluate the effectiveness of the proposed approach in comparison to existing SOTA EL tools, employing identical experimental parameters as prior comparative studies [2]. Notably, the entities are assumed to be pre-recognized (extracted). The study includes three datasets from the biomedical domain (see Table 1). For the main experiments, we employ Accuracy as the evaluation metric, representing the percentage of successful outcomes over the total number of input mentions. For the ablation studies, we utilize f-scores, precision, and recall. Our aim is to address the following research questions: **RQ1:** What is the efficacy of our neuro-symbolic methodology in biomedical entity linking tasks? **RQ2:** In what ways do domain-specific guidelines influence the effectiveness of *BioLinkerAI*? **RQ3:** How significant is the role of the sub-symbolic element within our approach?

4.1 Datasets

MedMentions encompasses 203,282 biomedical entities from 4392 abstracts of research articles. Both the validation and test sets contain entities present in the training data, as well as zero-shot entities (not encountered during training). The underlying KB is UMLS. Despite having a limited number of entities with definitions, it remains crucial to evaluate the impact of domain-specific expertise on the proposed neuro-symbolic approach. The experiments reuse the author-recommended ST21pv subset, where over 42 percent of test-time entities are zero-shot. Results reported are based on the test set. To ensure fair comparison with baselines, both seen and unseen configurations are employed. Given that the dataset includes entity types, the results of the experiments are reported in two scenarios: one (w/ Gold Types) where the entities’ semantic types are incorporated into the input to assist the study’s baseline methods in filtering candidates based on these semantic types, and another where the types are not considered.

The other dataset is BC5CDR ; it is another entity linking benchmark in the biomedical domain. The dataset comprises 1,500 PubMed articles’ abstracts. The dataset is annotated with entities from MeSH⁴ vocabulary. We used the mappings between UMLS and MeSH vocabulary to annotate the BC5CDR dataset with entities from UMLS knowledge base, as the current implementation of our approach links to UMLS knowledge base.

4.2 Baselines for Comparison

SAPBERT: achieves superior results compared to BioBERT [8] by introducing a self-alignment pretraining technique for large language models. **CLUSTERING-BASED**: employs an innovative clustering-based methodology to facilitate biomedical entity linking predictions [2]. **BIO-SYN**: applies a synonym marginalization approach for biomedical entity linking [19]. **INDEPENDENT**: crafted by [11], this entity linking model for zero-shot scenarios utilizes training on comprehensive descriptions of entities derived from expansive knowledge repositories. **SciSpacy**: utilizes the Spacy library for biomedical text processing [13]. **TF-idf**: represents a commonly employed candidate retrieval model [2, 14]. **DUAL**: enhances the entity linking process by integrating mention to mention coreference relationships [1]. **ARBORESCENCE**: enhances the clustering-based method by constructing minimum spanning arborescences across mentions and entities found within documents [1]. **BioEFG**: addresses biomedical EL challenges by using a generative adversarial network to produce latent features for rare entities. This enhances the encoder’s ability to accurately link entities in texts to KBs, especially for ambiguous mentions, demonstrating significant improvements on benchmarks [18]. [25]: enhances biomedical entity linking by employing a bi-encoder for initial retrieval

Table 4. illustrates example sentences from MedMentions with the expected entity; they show how the *BioLinkerAI* approach fails to link to the correct entity when we exclude the sub-symbolic part or the domain-specific rules.

Input	Gold Standard ID	Predicted IDs		
		wo/ sub-symbolic component	wo/ domain-specific rules	Neuro Symbolic
Interleukin 1 Beta	C0021753 (interleukin-1, beta)	-	C0021764(Interleukins), C0439096(Beta)	C0021753 (Interleukin 1 Beta)
These results indicated that nonylphenol diethoxylate has capability to affect	C2825142 (Experimental Result)	C1274040 (result)	-	C2825142 (Experimental Result)
Several studies reported somatic mutations of many genes (MEN1 , CTNNB1)	C0694884 (MEN1 gene)	C1704889 (MEN1 wt Allele)	-	C0694884 (MEN1 gene)
engagement with mental health services and management of chronic illness	(Disease Management)	C0376636 (Management procedure)	-	C0376636 (Disease Management)
an unscheduled return visit for additional treatment for persistent ...	C1512346 (Patient Visit)	C0545082 (Visit)	-	C1512346 (Patient Visit)

⁴ <https://www.ncbi.nlm.nih.gov/mesh/>.

and a prompt-learning-based encoder for context-aware re-ranking, effectively addressing term variety and ambiguity. **BioLinkerAI-Llama2**: to evaluate the model agnostic of our approach, we replaced GPT-4 with Llama2-70B [20] in the experiment study.

4.3 Main Results

Table 3 showcases our findings in comparison to baseline approaches for both the MedMentions and BC5CDR datasets. *BioLinkerAI* consistently outperforms or reports close performance compared to the studied baselines across both configurations (gold entity type and unseen/seen). In contrast to BERT-based approaches (SAPBERT and INDEPENDENT), which solely depend on training a language model using sentences and labeled entities, our neuro-symbolic approach does not need training over a new dataset (e.g., MedMentions); the sub-symbolic component (implemented on top of a LLM) is already trained over large collections of text data (e.g., books, articles, and web pages). This represents a crucial, foundational distinction our approach introduces to this field compared to baseline methods, regarding the pre-training process of LLMs and their subsequent utilization as knowledge repositories for inference. *BioLinkerAI-Llama2* performance is similar to *BioLinkerAI* (GPT-4) demonstrating the model-agnostic nature of our approach. *BioLinkerAI* demonstrates better performance compared to studied baseline over unseen data. BioEFG [18] achieves marginally better performance but relies entirely on language models that necessitate pre-training, incurring significant costs. In contrast, our method uses LLMs solely for ranking candidates. The usage of LLMs could be enhanced by incorporating additional rules that help determine the optimal instances to engage LLMs, making our approach more efficient and potentially more cost-effective. The results of this experiment successfully answers **RQ1**.

Ablation Studies Analyzing the Impact of Distinct Components: we present two setups: 1) *BioLinkerAI* excluding the sub-symbolic module "wo/" (excluding the LLM) by returning the top ranked candidate from the rule-based system (step 3 in Fig. 2) without applying step 4 in Fig. 2. 2) *BioLinkerAI* excluding domain-specific rules "wo/" by excluding the domain-specific rules (Rules 5, 6, and 7 in Fig. 2) from the approach; the generated candidates from step 2 in Fig. 2 are passed directly to the LLM (step 4 in Fig. 2) without ranking the candidates using the domain-specific rules. Table 2 shows how both components complement each other in performance (answering **RQ2** and **RQ3**). The incorporation of the sub-symbolic element substantially enhances *BioLinkerAI*'s efficacy. Eliminating domain-tailored guidelines leads to a decline in performance, as the symbolic module is not consistently capable of identifying the appropriate candidate. Examples in Table 4 illustrate this (e.g., the sub-symbolic component aids the *BioLinkerAI* approach to link "visit" by discarding entity (C0545082) labeled "Social Behavior", despite being an exact match, and selecting the appropriate entity (C1512346) labeled "Health Care Activity"). This demonstrates that in the medical domain, the integration of expert knowledge into a ML model is crucial, reinforcing the importance of the symbolic component.

5 Conclusion and Future Work

In this work, we introduced a neuro-symbolic framework for biomedical EL, composed of both symbolic and sub-symbolic components, demonstrating robust performance for both sentences and keywords. Our empirical findings indicate that the sub-symbolic system augments the effectiveness of the symbolic system, which employs human-crafted rule patterns. This integrated approach outperforms baselines rooted in neural networks. Our approach effectively eliminates the need for labeled training data when working with new datasets. Additionally, the interpretability of the system enabled us to trace errors at each stage of the process. Importantly, *BioLinkerAI* is flexible in its choice of components, allowing for substitution with alternatives that perform the same function (e.g., LLMs). Based on our findings, we encourage further exploration of the following research questions: 1) How can domain-specific rules be optimized for improved performance? 2) How can the neuro-symbolic approach be refined to deploy LLMs more selectively and efficiently?

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